2023 IEEE 8th International Conference for Convergence in Technology (I2CT)
Pune, India. Apr 7-9, 2023

Survey on Smartphone Sensors and User Intent in Smartphone Usage

Priyanka Bhatele
Department of Computer Engineering
Vishwakarma Institute of Technology
Pune, Maharastra, India
priyanka.bhatele@vit.edu

Mangesh Bedekar
School of Computer Engineering and Technology
MIT World Peace University
Pune, Maharashtra, India
mangesh.bedekar@mitpune.edu.in

2023 IEEE 8th International Conference for Convergence in Technology (I2CT) | 979-8-3503-3401-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/I2CT57861.2023.10126192

***Abstract*—Smartphone/Tablet users are approximately 3 million all over the world. It is likely to increase by several 100 million in the next few years. Around 40% of these users read online. Explicit means of feedback system is strongly based. It provides the most accuracy when rating an online learning application. Increase in the availability of content over the web and high user engagements, has led to the demand of the means that implicitly provide feedback. Implicit feedback relies on understanding the quality of the content based on the user activities performed over the web applications. Less accuracy is the limitation. It needs to stand with a support to provide as strong base as the explicit model does. Clipboard copy operations on the webpage provide an implicit insight to the user intentions. Screen activities like scrolling and pinch to zoom further can statistically be proven the positive indicators of user interest. Smartphone sensors like Gyroscope and Accelerometer silently sense human screen activities and mobile gestures. This review paper is based on the understanding of smartphone sensors and the inferences of user intent through it. The dig is based on various implicit indicators like mobile gestures, smartphone sensors and clipboard copy operations.**

Keywords—Smartphone sensors, Gyroscope, Accelerometer, Clipboard activity, Online learning platform

I. INTRODUCTION

Ubiquitous nature of the World Wide Web has made learning easy for the 21st century scholars. Availability of content online was in demand when the world has been hit by the pandemic.[39] Technology is changed in every 18 months of time. Earlier keyboard was the primary input device. Later in 1980s the computer mouse was introduced. Computer mouse are now gradually getting replaced by touch screen now. The progressive development of machine peripherals helps to understand the perception of the users handling them. Desktop systems has been an out-dated trend now. Tablets and Smartphones are the new choices among the youth for the purpose of e-learning. As per the study 64% students are willing to pay in the range of 5000-15000 for smartphones. [37] Millions of users are dwelling time on the e-learning application and websites. A fragment of JavaScript code is capable of inferring user behaviour over web by capturing the various activities performed over the web page. Sensors like Accelerometer and Gyroscope embedded into the Smartphones/Tablets to support the gaming industry [13]. These sensors calibrate silent monitoring activity of the users over the device [4, 14, 15]. Each tap on the mobile is associated with a tap print that has enough information on the interest and intentions of the user [17]. In smartphones, some other less accurate implicit indicators of web analytics are dwell time, scroll activity and pinch to zoom activity [2,3]. This paper is based to understand the context of the user behaviours through these implicit feedbacks and use them to interpret the intentions behind it.[40]

Different users have different purposes to access online available content. When an online content interests an end user, they will spend more time on the page [11]. The user will tend to perform some screen activities [2], mobile gestures [26] and copy activities [1, 24,16] as per intent of access. Figure 1 shows some of the usual activities performed by a user when trying to interestingly access an online content in smartphones/tablets.

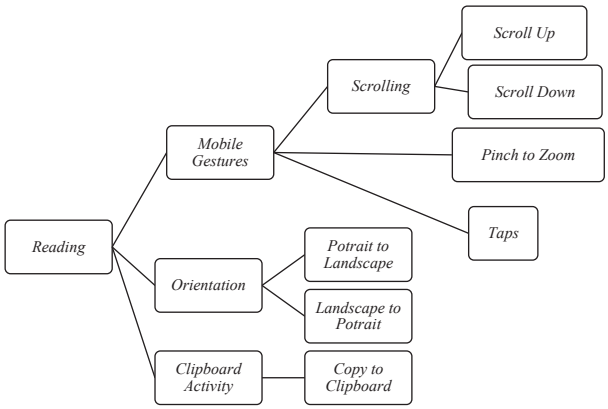

Fig. 1. User's usual activities when interestingly access an online content User while reading content online ought to perform some screen activities. Scrolling up and down for changing the viewport. Frequent taps to select. Pinch to zoom for better visibility. Orientation change from Landscape to Portrait or vice versa for comfortable reading. Copying text to clipboard for either more clarity or reference.

A. Mobile Gestures

A sequence of touch actions that happen in between the finger tap down on the screen to finger up movement.

1) Scrolling: Scrolling activity done on the screen of the smartphone to position the webpage in the viewport so that the content is visible and readable to the user.

2) Pinch to Zoom: Pinch-to-zoom is a common multi touch gesture that is done to zoom in and zoom out the viewport. The screen on the smartphones and tablets are smaller as compared to desktop system. Zooming in through the screen for better visibility is a common gesture for smartphone users. Moving two fingers apart on the touch screen is zoom in and bringing them together is zoom out.

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:54 UTC from IEEE Xplore. Restrictions apply.

3) Taps: A tap is a quick, gentle touch on the screen with a finger. Tapping once opens an app or selects an object. It is similar to a single left click using a mouse.

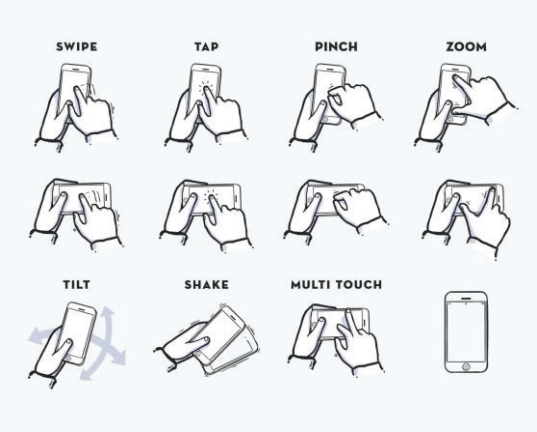

Fig. 2. Mobile Gestures
Some of the hand and finger movement while holding mobile for reading known as Mobile Gestures

B. Orientation

Physical orientation of the smartphones are detectable using the gravity felt on its axis. When a device is rotated in the portrait condition, gravity is felt at the y-axis. When it is in the landscape mode, gravity is felt at x-axis.

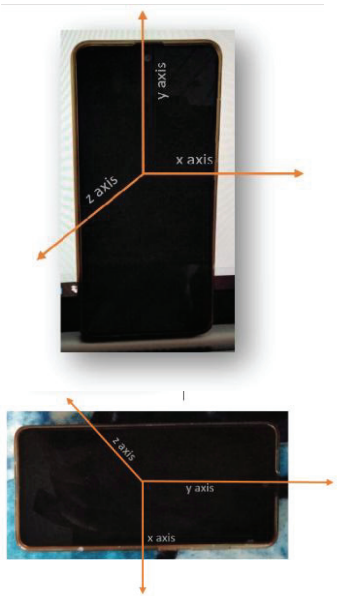

Fig. 3. Landscape and Portrait Orientation of a Smartphone Device
Plot of x-y-z axis and the gravity effects for smartphone device in an orientation.

C. Clipboard Activity

Content copied in any system is saved in the clipboard. This utility is useful because saving the copied content can then be pasted multiple times. The content is forgotten by the system only when the new content is copied.

This research paper is dedicated to understand the study of the smartphone sensors and its varied application. Another proposed approach of the manuscript is to use the overabundance of data collected from these sensors to implicate user intent behind the smartphone usage. Section II briefs the entire literature surveyed related to the proposed objective. Section III summarizes the survey. Section IV presents the investigation done in order to understand smartphone sensors readings during mobile gestures. Section V shows some results followed by conclusion and future work.

II. Literature Review

The survey has been categorised in following parts: - Implicit feedback, Smartphone usage, Smartphone sensors and its current research explorations, webpage user activities, derivatives using webpage user activities and Mobile Gestures.

A. Implicit Feedback

The complete recommender system relies on feedback. Suggestions made by such systems are based on the deep understanding of the interest and intent of the user. Recommender systems that determine user interests by explicit ratings are Explicit Feedback systems. It explicitly asks user for input and user is proactively provides ratings. On the contrary those systems that intelligently determine the interests of the user are Implicit Feedback systems. It is based on the indirect ways of understanding the satisfaction level of users. [2,38] Mark Claypool et. al. analysed the relationship between the various user actions on a web page with the explicit feedback. Some of the user actions were found to be in correlation with the explicit ratings. A web browser "The Curious Browser", developed by the team, is capable of capturing implicit indicators while web browsing. The browser collected web browsing data for 40 hours by 70 students. The graphical user interface also notified users to provide the ratings of the web pages explicitly. The implicit indicators taken under consideration for the experiment are: -

1) *Dwell time*
2) *Mouse activities*
3) *Scrolling activity*
4) *Keyboard activity*

Kruksal-Wallis test was done to examine the degree of independence for each implicit indicator mentioned above. Table 1 shows the results of the analysis:

TABLE I. Statistical Correlation Analysis of Implicit Indicators with Explicit Interests

S.no.	Implicit Indicator	Result Analysis
1	Dwell time	Strong correlation
2	Mouse activities	Ineffective
3	Scrolling activity	Strong correlation
4	Keyboard activity	Strong correlation
5	Dwell time & Scrolling activity combined	Strong correlation

These implicit indicators were found to be 70% accurate by Mark Claypool et. al. [2] when compared against explicit feedback recorded from the user.[2] Despite being accurate enough, implicit feedback methods are considered to be less reliable as explicit methods are being indirect. They can be either combined with an explicit model for enhancement or combined with other implicit methods for better accuracy.

Google Analytics is a tool that provides an effective data analysis over a web page. Incorporated by Google, the software is open source and available to users. Heng et.al. [8] have studied Google Analytics over some research questions like: -

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:54 UTC from IEEE Xplore. Restrictions apply.

1) User identification
2) Patterns and trends in learning
3) Technologies used to access the course
4) Strengths and Limitations

Google Analytics was found to answer most of them. The tool is effective in providing the overview of the aggregated statistics of sessions, views, dwell time and user details. Data visualization is the predominant strength of the tool. However, it is not efficient in terms of providing some implicit details about the user. The tool also found to be ineffective in collecting personalized learning data.

Viewport is that part of any content on the web page which is visible to the user at a time on the device. Time analysis of the viewport for the online new reading is done with a mixture of Markov chains.

Dmitry et. al. [11] studied the depth of the article consumption. The study defines four level categories of attention amongst the users: -

1) Bounce - User did not engage at all and leaves the article quite rapidly.
2) Shallow Engagement – User engaged < 50% into the article.
3) Deep Engagement – User engaged > 50% into the article
4) Complete Engagement – User has completely read the article and has reached the comment section to provide his views.

B. *Smartphone Usage*

Qiang Li et. al. designed an application UserIntent that understands user intent by the smartphone sensors readings and launches the required smartphone application. User creates their own gestures.[34]

Yangil Park et. al. explored the acceptance of smartphone for innovations. The investigations results that the perceived usefulness (PU) and perceived ease of use positively (PEOU) completely determines the human motivation towards usage of smartphone among medical professionals.[35]

Dong-Hee Shin et. al. studied the adoption of smartphones as a pedagogical tool for learning digitally. The major subjects for the study included the age group of 21-30 yrs. Under graduates, graduates and doctorates were the predominant degrees chosen for the research. The results shown high level of satisfaction among the subjects on the smartphone as an effective tool for learning.[36]

S. Saraswathi has studied the popularity of smartphones among students. The survey included the students of age group 19-25yrs. predominantly. Smartphones were found to be common for downloading study materials, and taking pic of lectures and notes. Logging in to the academic work were another general task that were popular among students using smartphones. 68% of the students opinionated that smartphone is a useful device for education.[37]

Young scholars of age groups 6-12yrs, are also focussing on the learning digitally through mobile devices.[39] Another investigation on the young children ranging the age group from birth to 5yrs was done by Xie H et. al. The study was based on the learning capabilities using touchscreen devices. Children using touchscreen devices performed better as compared to the ones not using it.[41]

C. *Smartphone Sensors*

1) *Accelerometer and Gyroscope*

Sensors like Accelerometer and Gyroscope is embedded into most of the smartphones to support gaming industry. User screen activities like tap on the screen, zoom activity, orientation of the mobile etc. is sensed in a continuous manner by these sensors. [13].

Accelerometer measures the acceleration experienced by the device. Gyroscope makes use of the principle of Angular Momentum. It helps changing the orientation by detecting the spin movement. Unlike GPS, accelerometer and gyroscopic sensors do not require permission from the user to capture data. Fig.3 shows a sample reading of accelerometer during a tap on the device.Similar readings are generated by the gyroscope sensor [13, 14]. These sensors have been successfully used for biofeedback

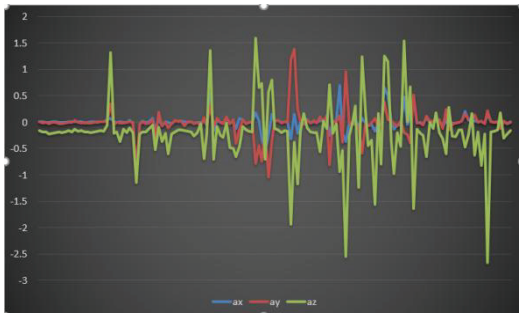

Fig. 4. Accelerometer Sensor Reading during a tap on the device
Plot of the sample readings generated from the accelerometer embedded in smartphone device. Blue lines represent x-axis, red line y-axis, green lines z-axis

2) *Biofeedback*

Biofeedback is one of the recent applications of Mobile Sensors. Human activities can be recognized and monitored in an accurately without placing any external device on the human body. Smartphones are capable of capturing data from the sensors like accelerometer and gyroscope on the human activities like walking, standing, running, climbing staircases etc. [4, 13, 14, 15,19].

Robert et.al. [14] created an application that captured data from accelerometer and gyroscope sensors. Huge dataset was gathered on activities like walking, sitting, standing, running etc. Classification algorithm was applied to this dataset to create machine learning model. Any new data from the sensors can be applied to this model to get the result on the human activity performed. The model provides accuracy of 92% and 73% on the dataset created by the author and the dataset available by other researchers. The data set is made public by the researchers and can be utilized for future use.

Muhammad Ehatisham-ul-Haq et. al. applied various machine learning algorithms mentioned below on data generated from sensors: -

1) Decision Tree
2) K-NN
3) Bayes Net
4) Support Vector Machine Algorithm (SVM)

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:54 UTC from IEEE Xplore. Restrictions apply.

Among the above, SVM found to perform the best over others with an accuracy of 99.18% [15].

3) TapPrints

Emiliano Miluzzo et. al. [17] have given a very logical name to the finger tap on a smartphone/tablet as TapPrint. A TapPrint of a finger tap on the device was a result of classification model that is based on Ensemble Learning. The dataset after feature extraction was passed to different classification algorithms and the final result was predicted based on the majority. The data collection for this model was based on the following operations: -

1) Icon Tap
2) Sequential Letters
3) Pangrams
4) Repeated Pangrams

The experiments were carried out on both a Smartphone and a Tablet.

4) User Intent and Context Recognition

An application was developed by T Ilmjärv et al. [25] to capture readings from gyroscope and accelerometer while a person is reading in smartphone. Data labelling was supported with the image captured simultaneously of the user. Neural network, Decision Tree and K-NN algorithm were considered to create the learning model. Considering the complexity of the features, neural network was implemented. Classification of the reading activity over mobile using neural network achieved the accuracy of 94%. This classification can be used to precise the mobile content usage statistics that can be further considered for various inferences [26].

With respect to the goals of user search, Christoph Kofler et. al. has suggested two dimensions, topical (what) and intent (why). User intent is defined as the purpose, goal or the immediate reason behind any search. The textual information does not always support the complete information behind the search. User intent is the key factor for the accurate query results.[42]

Context recognition is the automatic inference of the circumstantial information about the user's situation. Context Data Collector (CDC), an application was developed by Abayomi Moradeyo Otebolaku et. al., to collect built-in smartphone sensors readings for the physical activities like sitting, standing, driving, jogging, and walking. List of algorithms k-NN, Decision Trees, K-Star, SVM, MLP Logistic Regression, RIPPER, Naïve Bayesian Network and PART were assessed with the data collected. High performer classifiers found out to be Random Forest , k-NN and MLP, with 98.67% , 98.77% and 98.1% respective accuracies.[40]

Smartphone sensors are capable of collecting plethora of data in continuous manner that can be sampled and used to understand the user implicit behaviour. These factors when mapped to the intentions behind may revolutionize several sectors and prove to be a boon to society. [3]

5) Implicit Authentication

Rehzhong Wang et. al. worked on enhancing the pin authentication by applying One-Class SVM classifier to the 1000+ smartphone sensor date of 87 participants. The data was examined in both static and dynamic scenarios. The classifier outperformed with the equal error rate of 0.0071%.[30]

Jiyun Wu et. al. exploited the fact that an unauthorized access to smartphones has changed keystroke behaviour. User identity authentication algorithm was developed done on the password length of 8 digits using SVM classifier. Various features were taken into account and the classifier performed with the EER of 1.2514%.[31]

6) Multimedia Smartphone Sensor

Various work has been done using the multimedia sensors embedded in the smartphones. Lorena Para et.al. developed a system that uses both microphone and camera to provide ambient assisted living and e-health.[33]

D. Webpage User Activity

1) Copy to clipboard

An extension of javascript code can be applied on the web browser to capture the copy activity of the users directly from the clipboard. Collection of this copy to clipboard data can be used for the following purpose [1, 24, 16]: -

1) Text Summarization
2) Text Simplifications
3) Understand Audience's Interest
4) Tracking User progress on Tutorials
5) Search Engine Optimization
6) Language Translations Required
7) Complex Word Identifications

Heatmaps are an effective way of visualizing the data graphically copied to the clipboard by the user. Different shades of warm and cold colours are used to represent different frequency level of copying. Such heatmaps provides a better qualitative understanding on the online learning applications. Another vital application of heatmaps in the study of user intent could be identification of complex words. The more frequently copied words are more complex [16, 21, 24].

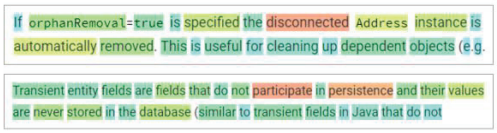

Fig. 5. Sample of heatmaps based on the copy to clipboard data [16]
Heatmaps represents data in the form of colours. The frequency ranges are shown in colours. Warm colours like red and orange shows high frequency of copied data. Green, blue and yellow are less frequently copied data.

2) Horizontal Mouse Movements

Human being performs finger assisted reading when reading a book or offline content. Finger is replaced by the Mouse cursor when it comes to reading content online. Mouse cursor can be correlated with the eye gaze of the user. User less likely moves out of the page after a Left to Right horizontal move on the web page. Kirsh and Joy have researched successfully over the mouse activity parameters mentioned below, that Horizontal Mouse Movements (HMMs) can be considered as an indicator to user interest [23]: -

1) Horizontal Distance
2) Horizontal Direction
3) Vertical range

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:54 UTC from IEEE Xplore. Restrictions apply.

4) Time Frame

Pointer assisted reading is done by the user when reading the content online. User is likely to move the cursor slowly towards the right than to the left. Cursor movement to the left is a quick movement and is mostly done to switch between the lines or web pages. Horizontal mouse movement are more frequent as compared to vertical. Smartphones and Tablets do not have a mouse attached, and hence capturing of reading pattern is an open gap in those devices [20].

E. Derivatives

1) Complex Word Identification (CWI)

User copy operations can be effectively used to identify complex word among the web content. Words having more syllable count, more characters and less frequency are likely to be complex than others. Web users mostly copy such words to better understand the meaning separately from Google translators. Kirsh has introduced a new approach taking Syllable count, Character length and Frequency of words as parameters [12]. This approach automatically identifies complex words on a web page and can be further user for text simplification. Simplified text will increase the count of audience on the web page.

2) Extractive Text Summarization: - HCI Approach

Manual Summarization of text can be tedious tasks. Extractive methods select key sentences from the complete content based on what is mostly copied from the user. This method calculates the score of every sentence and includes them into the summary. It filters the data by only the considering sentences including word range 8 to 40. Identified sentences having the count above a threshold value are included into the summary without any change [6].

3) Sub-page Metrics & KPI'S

Maintaining, improving and optimizing activities of a web page is done using some Page Metrics and Key performance Indicators. These metrics consider web page as an indivisible atom. Kirsh and Joy have defined sub-page metrics and KPIs that takes paragraphs and sub part of the pages as the potential metrics and Key Performance Indicators [10]. A positive indicator could be a fragment of code of an online learning website page that is frequently copied by the users. A negative indicator could be some complex words that are copied by the users frequently for further clarity. These studies are based on the cursor activity done by the user. Defining sub-page metrics and KPI's is an open-ended gap in Smartphones and Tablets.

F. Mobile Gestures

1) Gesture Recognition

In smartphones and tablets various mobile gestures like swiping, scrolling and pinch to zoom activity can be used to understand the context of the user actions. A mobile gesture is a sequence of touch actions that happen in between the finger tap down on the screen to finger up movement. [7, 3, 25]

Machine learning algorithms can be incorporated to identify such mobile gestures and further help understanding the intent of the user while performing an activity in smartphone devices.

Various hand and mobile sensors have been proposed by Kirsh et.al [27] to act as a phone pointing remote. The proposed model will understand the hand gestures without using the mobile touch screen and get inference from the values of the smartphone sensors. These gestures when completely mapped to perform device operations in the IoT devices like television, air conditioner etc.

Mohammed A. Alqarni et. al. designed a sensor-based authentication algorithm. The machine learning algorithms (SVM, Random Forest and Bayes net) identifies user based on the mobile interaction using smartphone sensors readings. The implicit authentication method can be combined along with the explicit authentication methods (pin login, password protection, fingerprints, face recognition etc.) to enhance the verification of the mobile users. Study of 12 hand gestures and mobile gestures by 26 users was done. Among the SVM, RF, and BN classifiers, RF classifier performed best for both hand and mobile gestures with an average accuracy of 74.97%. [29]

2) Surface Gestures

Lisa Anthony et. al. explored the benefit of visual feedback for surface gestures. The prime subject for this study was pre teenagers and teenagers, though participants also included some adults. A set of 20 gestures were asked to be drawn on the screen of a smartphone with and without visual feedback. The gestures included some letters, numbers, signs and basic shapes. ANNOVA test was conducted on the simple features (dimension, pressure and speed) and complex features (orientation, angle and sharpness) of the gestures. The study concluded that the visual feedback is a requirement for the children in contrast to the adults.[28]

G. Online Learning

1) u-learning

Ubiquitous learning is the prevalence of learning materials and education. The learning methods includes learning that includes learning tools as smartphones, laptops and other mobile devices.[36]

2) e-learning

e learning, referred as electronic learning, is defined as the availability of teaching content with media in Internet or computer network media.[32]

3) m-learning

Mobile learning is the learning that involves usage of mobile devices, a small device with limited computing abilities. Referring of courses and materials anytime and anywhere. This kind of learning involves the mobile communications technologies like smartphones.[32]

III. Summary and Comparative Analysis of all the Literature

Table 2 shows the comparative analysis of various implicit methods. Most of the implicit indicators (page views, dwell time, mouse activity, scrolling activity and keyboard activity) are based on the peripheral device activities for desktop systems. Smartphone devices still lack to have implicit indicators. Clipboard activity and mobile gestures may lead to be useful methods to perceive implicit interests of smartphone usage.

Table 3 summarizes the consolidated study of the selection of age groups for the various studies among scholars. The summary illustrates the age group 0-45 yrs. has been considered for most of the research. The age group of 45 above remains unexplored.

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:54 UTC from IEEE Xplore. Restrictions apply.

Table 4 summarizes the usage of all the classifiers for context identification through activities performed, using smartphone sensors readings. The performance details that Support Vector Machine has been the foremost choice for classifying the sensor readings, with the highest accuracy. Most of the dataset is labelled and hence supervised learning methods are used for all the studies. Accelerometer and Gyroscope is the foremost smartphone sensor used by all the researches.

TABLE II. COMPARATIVE ANALYSIS OF IMPLICIT FEEDBACK METHODS

Study	Data Extraction Means	Page Views	Dwell Time	Mouse Activity	Scrolling Activity	Keyboard Activity	Clipboard Activity	Mobile Gestures
The Curious Browser [2]	Application	✓	✓	✓	✓	✓		
Google Analytics [8]	Application	✓	✓					
Viewport Analysis [11]	Javascript Code	✓	✓		✓			
Word Attention Heatmaps [16]	Javascript Code	✓		✓			✓	
Horizontal Mouse Movements [23]	Javascript Code	✓		✓				
Swipe and Tell [4]	Application	✓	✓					✓

TABLE III. CONSOLIDATED SUMMARY OF THE AGE GROUP SELECTION FOR THE STUDY OF SMARTPHONE USAGE AMONG SCHOLARS

Study	Up to 5 yrs.	5-10 yrs.	10-18 yrs.	18-25 yrs.	25-35 yrs.	35-45 yrs.
Xie H et. al. [41]	✓					
Mohd Zdikri Md Sabron et. al. [39]		✓	✓	✓		
Dong-Hee Shin [36]				✓	✓	✓
S. Saraswathi [37]			✓	✓	✓	

TABLE IV. COMPARATIVE ANALYSIS FOR PERFORMANCE OF CLASSIFIERS USING SMARTPHONE SENSORS READINGS

Study	Dataset	Sensors Used	Activities	Context	Scenario	Classifier along with Performance	Accuracy
Robert Andrei Voicu [14]	15,750	Accelerometer and Gyroscope	Walking Running Sitting Standing Upstairs Downstairs	Biofeedback	Static and Dynamic	Multi-layer Perceptron (MLP)	93%
Muhammad Ehatisham-ul-Haq [15]	10,800	Accelerometer and Gyroscope	Walking Standing Running Sitting Walking Upstairs Walking Downstairs	Implicit Authentication	Static and Dynamic	Decision Tree	96.82
						K-Nearest Neighbor	93.30
						Bayes Net	97.38
						Support Vector Machine	99.18
Emiliano Miluzzo [17]	40, 000	Accelerometer and Gyroscope	Tap Icon Tap Sequential Letters Pangrams Repeated Pangrams	Implicit Authentication	Static	Ensemble Classification (Bagged Decision Trees, Random Forests, Support Vector Machines, k-NN, Multinomial Logistic Regression)	90%
Taavi Ilmjarv [26]	5000	Accelerometer and Gyroscope	Reading	Context Aware	Static	Neural Network	94%
Abayomi Moradeyo Otebolaku et. [40]	-	GPS, light, microphone Accelerometer and Gyroscope	walking, jogging, lying, running, standing, driving, sitting. ascending stairs, descending stairs, ascending in elevator, and descending in elevator	Context Aware	Static and Dynamic	k-NN	98.77%
						Random forest	98.67%
						MLP	98.1%

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:54 UTC from IEEE Xplore. Restrictions apply.

Study	Dataset	Sensors Used	Activities	Context	Scenario	Classifier along with Performance	Accuracy
Renzhong Wang et. al. [30]	10000+	accelerometer, gyroscope, magnetometer, pressure, touch size	Mobile Security Pin	Implicit authentication	Static and Dynamic	One-Class SVM	equal error rate (EER) - 0.0071%
Jiyun Wu et.al [30]	-	accelerometer, gyroscope, magnetometer, pressure, touch size	Mobile Security Pin	Implicit authentication	Static	Support Vector Machine (SVM)	99.1329%
Kirsh et.al. [27]	-	accelerometer, gyroscope,	Hand Gestures without using Screen	Gesture Recognition	Dynamic	Proposed, yet to be Implemented	
Mohammed A.Alqarni et.al. [29]	-	accelerometer, gyroscope	Hand and Mobile gestures	Gesture Recognition	Dynamic	Support vector machine	74.78%
						Bayes net	64.38%
						random forests	74.97%

IV. Understanding Gyroscope and Accelerometer Sensors with Smartphone handling

Understanding of smartphones sensors (gyroscope and accelerometer) with respect to some mobile gesture is mandate. In order to accomplish the same, an analysis has been done on the readings captured while performing below activities:

1) Keeping the smartphone in Idle state
2) Orientation change (Landscape to Portrait and vice versa)
3) Pinch to zoom
4) Scrolling over screen
5) Tap on the screen
6) Rotation along x, y and z axis (yaw, pitch, roll) [43]

The readings from accelerometer and gyroscope were captured using the application developed *"Books"*. The application captures the data in the csv format. Fig. 6 shows sample readings generated.

Date	timestamp	gx	gy	gz	ax	ay	az
07-09-2022	17:23:00	0.539688	0.824037	-0.7667	-0.03495	2.50995	6.49395
07-09-2022	17:29:14	-0.01801	-0.03905	-0.00701	-0.684	1.25595	9.672001
07-09-2022	17:31:32	0.023512	0.055138	-0.06848	9.927	1.32	3.042
07-09-2022	17:35:06	-0.0143	-0.1122	-0.03438	-2.67105	3.95595	8.556001
07-09-2022	17:23:00	-0.36809	-0.67499	0.3146	-1.31595	3.95895	8.467051

Fig. 6. Sample readings from gyroscope and accelerometer using "Books" application
gx-gy-gz: readings from gyroscope
ax-ay-az readings from accelerometer

V. Results and Discussion

Fig. 7a-h demonstrates plotted line graphs against readings for accelerometer. Different pattern can be observed for each mobile gesture. This data can be used to train a machine learning models for any further application in research.

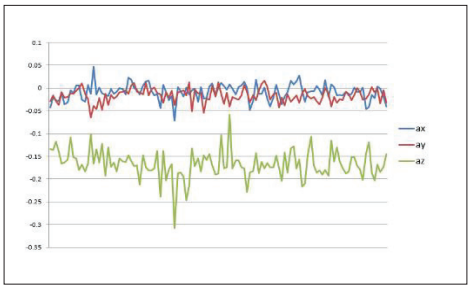

Fig. 7a: Activity Type: Smartphone kept Idle

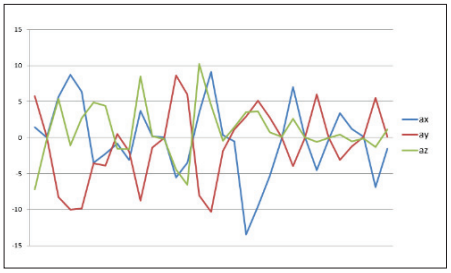

Fig. 7b: Activity Type: Orientation change (Landscape to Portrait and vice versa)

Fig. 7c: Activity Type: Pinch to zoom

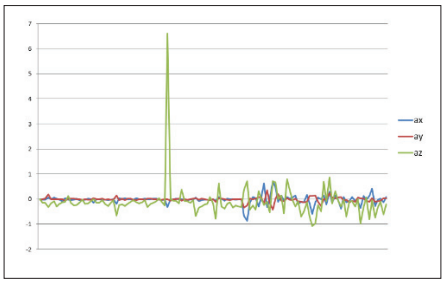

Fig. 7d: Activity Type: Scrolling over screen

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:54 UTC from IEEE Xplore. Restrictions apply.

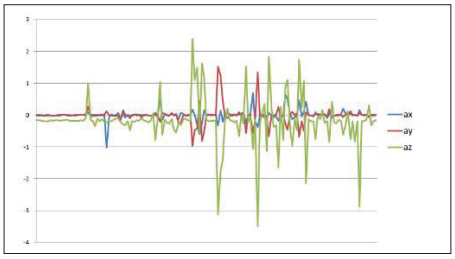

Fig. 7e: Activity Type: Tap on the screen

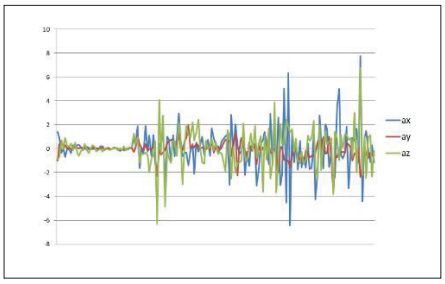

Fig. 7f: Activity Type: Rotation along x-axis (pitch)

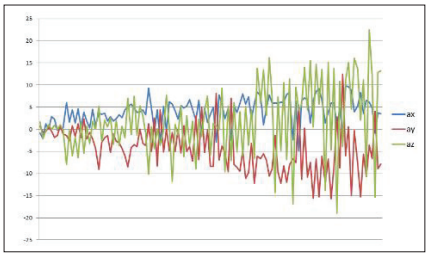

Fig. 7g: Activity Type: Rotation along y-axis (yaw)

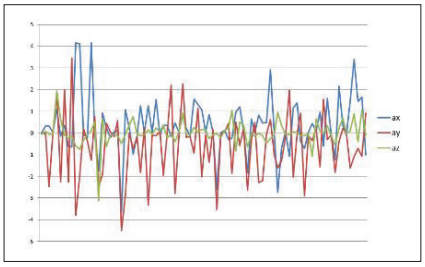

Fig. 7h: Activity Type: Rotation along z-axis (roll)

VI. Concluding Remarks and Future Work

With the increase in the technological era, millions of users own a Smartphone device. Online learning application and websites have seen an increase in the number of users dwelling time over it. Various research works has been done on the web applications running over the desktop systems. Near about 67.2% of the world's population is engaged in handheld devices. Smartphones/Tablets have proven to be the new classrooms for the students. Reading and referring content online has expediency over traditional methods. Improving the content over these platforms and optimizing it has been the requirement and an open area of research. This review paper based its exploration in smartphone sensors and clipboard activities to infer user intent in smartphone usage. User performs some mobile gestures, screen orientation change and clipboard activities while reading using smartphones/tablets. Implicit indicators, when measured independently, has limitations on accuracy. It is not capable enough to standalone against the explicit feedback system. Plethora of data sensed by smartphone sensors has been used as an enhancement of device authentication, biofeedback and other context aware. Readings from the sensors embedded into smartphones has the capability of bring implicit indicators. Various classifiers have been applied to classify these readings. It has the capability to be the base for understanding the user intent and behaviour. On the contrary, such standalone feedback methods always has a limitation on accuracy. Combination of more than one implicit method of feedback can be the solution. Comparing it with the explicit feedback systems remains unexplored. Copy activity of user has been studied for the desktop devices. Frequent copying activity hypothetically relates to implicit feedback of user's interest. By combining the independent implicit feedbacks from the user screen activities, mobile gestures and copy activity, a standalone multimodal system can be designed that will act as an accurate feedback system. Future review and investigation be done on the accuracy of the smartphone sensors. Study related to the application of reinforcement learning on the dataset collected from smartphone sensors can be done.

References

[1] Kirsh, I. (2020). What Web Users Copy to the Clipboard on a Website: A Case Study, WEBIST, In Proceedings of the 16th International Conference on Web Information Systems and Technologies (WEBIST 2020), SciTePress, INSTICC, 2020. August 2020 http://dx.doi.org/10.5220/0010113203030312

[2] Mark Claypool, Phong Le, Makoto Wased, and David Brown. 2001. Implicit interest indicators. In Proceedings of the 6th international conference on Intelligent user interfaces (IUI '01). Association for Computing Machinery, New York, NY, USA, pp. 33–40. https://doi.org/10.1145/359784.359836

[3] KlaasNelissen, Monique Snoeck, SeppeVandenBroucke, and Bart Baesens. 2018. Swipe and Tell: Using Implicit Feedback to Predict User Engagement on Tablets. ACM Trans. Inf. Syst. 36, 4, Article 35 (October 2018). https://doi.org/10.1145/3185153

[4] Umek, A., Kos, A. Validation of smartphone gyroscopes for mobile biofeedback applications. Personalised and Ubiquitous Computing 20, pp. 657–666 (2016). https://doi.org/10.1007/s00779-016-0946-4

[5] Sarun Sumriddetchkajorn and Armote Somboonkaew "Low-cost cell-phone-based digital lux meter", Proc. SPIE 7853, Advanced Sensor Systems and Applications IV, 78530L (9 November 2010). http://dx.doi.org/10.1117/12.870176

[6] Kirsh, I. and Joy, M. (2020c). An HCI approach to extractive text summarization: Selecting key sentences based on user copy operations. In Proceedings of the 22nd HCI International Conference (HCII 2020), Communications in Computer and Information Science, Cham. Springer International Publishing. https://doi.org/10.1007/978-3-030-60700-5_53

[7] Y. Li, "Beyond Pinch and Flick: Enriching Mobile Gesture Interaction" in Computer, vol. 42, no. 12, pp. 87-89, 2009. https://doi.org/10.1109/MC.2009.381

[8] Luo, Heng & Rocco, Stevie &Schaad, Carl. (2015). Using Google Analytics to Understand Online Learning: A Case Study of a Graduate-Level Online Course, pp. 264-268. https://doi.org/10.1109/EITT.2015.62.

[9] Kirsh, I. (2020). Directions and Speeds of Mouse Movements on a Website and Reading Patterns: A Web Usage Mining Case Study. Proceedings of the 10th International Conference on Web Intelligence, Mining and Semantics. June 2020, Pp 129–138. https://doi.org/10.1145/3405962.3405982

[10] Kirsh, I., & Joy, M. (2020). Splitting the Web Analytics Atom: From Page Metrics and KPIs to Sub-Page Metrics and KPIs. Proceedings of the 10th International Conference on Web Intelligence, Mining and Semantics. June 2020, Pages 33–43, https://doi.org/10.1145/3405962.3405984

[11] Dmitry Lagun and Mounia Lalmas. 2016. Understanding User Attention and Engagement in Online News Reading. In Proceedings of the Ninth ACM International Conference on Web Search and Data Mining (WSDM '16), Association for Computing Machinery, New York, NY, USA, pp. 113–122. https://doi.org/10.1145/2835776.2835833

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:54 UTC from IEEE Xplore. Restrictions apply.

[12] Kirsh, I. (2020a). Automatic complex word identification using implicit feedback from user copy operations. In Proceedings of the 21st International Conference on Web Information Systems Engineering (WISE 2020), Lecture Notes in Computer Science, forthcoming, Cham. Springer International Publishing. https://doi.org/10.1007/978-3-030-62008-0_11

[13] Liu, M. (2013). A Study of Mobile Sensing Using Smartphones. International Journal of Distributed Sensor Networks. https://doi.org/10.1155/2013/272916

[14] Voicu R-A, Dobre C, Bajenaru L, Ciobanu R-I. Human Physical Activity Recognition Using Smartphone Sensors. Sensors. 2019; 19(3):458. https://doi.org/10.3390/s19030458

[15] Ehatisham-ul-Haq M, Azam MA, Loo J, Shuang K, Islam S, Naeem U, Amin Y. Authentication of Smartphone Users Based on Activity Recognition and Mobile Sensing. Sensors. 2017; 17(9):2043. https://doi.org/10.3390/s17092043

[16] Kirsh, I. and Joy, M. (2020a). A different web analytics perspective through copy to clipboard heatmaps. In Proceedings of the 20th International Conference on Web Engineering (ICWE 2020), Lecture Notes in Computer Science, vol 12128, pp. 543–546, Cham. Springer International Publishing. http://doi.org/10.1007/978-3-030-50578-3_41

[17] Emiliano Miluzzo, Alexander Varshavsky, Suhri d Balakrishnan, and Romit Roy Choudhury. 2012. Tapprints: your finger taps have fingerprints. In Proceedings of the 10th international conference on Mobile systems, applications, and services (MobiSys '12). Association for Computing Machinery, New York, NY, USA, pp. 323–336. https://doi.org/10.1145/2307636.2307666

[18] Luis A. Leiva, Ioannis Arapakis, and Costas Iordanou. 2021. My Mouse, My Rules: Privacy Issues of Behavioral User Profiling via Mouse Tracking. In Proceedings of the 2021 Conference on Human Information Interaction and Retrieval (CHIIR '21). Association for Computing Machinery, New York, NY, USA, pp. 51–61. https://doi.org/10.1145/3406522.3446011

[19] Liang Cai, Sridhar Machiraju, and Hao Chen. 2009. Defending against sensor-sniffing attacks on mobile phones. In Proceedings of the 1st ACM workshop on Networking, systems, and applications for mobile handhelds (MobiHeld '09). Association for Computing Machinery, New York, NY, USA, pp. 31–36. https://doi.org/10.1145/1592606.1592614

[20] Kirsh, I. and Joy, M. (2020b). Exploring Pointer Assisted Reading (PAR): Using mouse movements to analyze web users' reading behaviors and patterns. In Proceedings of the 22nd HCI International Conference (HCII 2020), Lecture Notes in Computer Science, Cham. Springer International Publishing. https://doi.org/10.1007/978-3-030-60117-1

[21] Kirsh I. (2021) Visualizing Web Users' Attention to Text with Selection Heatmaps. In: Brambilla M., Chbeir R., Frasincar F., Manolescu I. (eds) Web Engineering. ICWE 2021. Lecture Notes in Computer Science, vol 12706. Springer, Cham. https://doi.org/10.1007/978-3-030-74296-6_42N .

[22] N. D. Lane, E. Miluzzo, H. Lu, D. Peebles, T. Choudhury and A. T. Campbell, "A survey of mobile phone sensing," in IEEE Communications Magazine, vol. 48, no. 9, pp. 140-150, Sept. 2010, https://doi.org/10.1109/MCOM.2010.5560598

[23] Kirsh, I., Joy, M., &Kirsh, Y. (2020). Horizontal Mouse Movements (HMMs) on Web Pages as Indicators of User Interest. (HCI International 2020), CCIS, Springer, 2020. https://doi.org/10.1007/978-3-030-60700-5_53

[24] Kirsh, I. (2020b). Using mouse movement heatmaps to visualize user attention to words. In Proceedings of the NordiCHI '20, Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society, October 2020 Article No.: 117, Pp. 1–5. https://doi.org/10.1145/3419249.3421250

[25] Qi Guo, Haojian Jin, Dmitry Lagun, Shuai Yuan, and Eugene Agichtein. 2013. Mining touch interaction data on mobile devices to predict web search result relevance. In Proceedings of the 36th international ACM SIGIR conference on Research and development in information retrieval (SIGIR '13). Association for Computing Machinery, New York, NY, USA, 153–162. https://doi.org/10.1145/2484028.2484100

[26] T. Ilmjarv et al., "Detecting user reading behaviour using smartphone sensors", 2015. Bachelor Thesis, University of Tartu, Faculty of Mathematics and Computer Science, Institute of Computer Science. https://core.ac.uk/download/pdf/79113289.pdf

[27] Kirsh, I., Ruser, H.: Phone-pointing remote app: Using smartphones as pointers in gesture-based IoT remote controls. In: Proceedings of 23rd HCI International Conference (HCII'21), CCIS 1420, Springer (2021). https://doi.org/10.1007/978-3-030-78642-7_3

[28] Lisa Anthony, Quincy Brown, Jaye Nias, Berthel Tate, Children (and adults) benefit from visual feedback during gesture interaction on mobile touchscreen devices, International Journal of Child-Computer Interaction, Volume 6, 2015, Pages 17-27, ISSN 2212-8689. https://doi.org/10.1016/j.ijcci.2016.01.002.

[29] Alqarni, M.A., Chauhdary, S.H., Malik, M.N. et al. Identifying smartphone users based on how they interact with their phones. Hum. Cent. Comput. Inf. Sci. 10, 7 (2020). https://doi.org/10.1186/s13673-020-0212-7

[30] R. Wang and D. Tao, "Context-Aware Implicit Authentication of Smartphone Users Based on Multi-Sensor Behavior," in IEEE Access, vol. 7, pp. 119654-119667, 2019, https://doi: 10.1109/ACCESS.2019.2936034

[31] Wu J, Chen Z. An Implicit Identity Authentication System Considering Changes of Gesture Based on Keystroke Behaviors. International Journal of Distributed Sensor Networks. June 2015. https://doi:10.1155/2015/470274

[32] Lu'mu 2017 Learning Media Of Applications Design Based Android Mobile Smartphone Int. J. Appl. Eng 12 pp 6576-6585.

[33] Parra, L., Sendra, S., Jiménez, J.M. et al. Multimedia sensors embedded in smartphones for ambient assisted living and e-health. Multimed Tools Appl 75, 13271–13297 (2016). https://doi.org/10.1007/s11042-015-2745-8

[34] Li, Qiang & Han, Qi & Sun, Limin. (2014). UserIntent: Detection of user intent for triggering smartphone sensing applications. 185-187. https://doi.org/10.1109/SAHCN.2014.6990351

[35] Park, Y. and Chen, J.V. (2007), "Acceptance and adoption of the innovative use of smartphone", Industrial Management & Data Systems, Vol. 107 No. 9, pp. 1349-1365. https://doi.org/10.1108/02635570710834009

[36] Dong-Hee Shin, Youn-Joo Shin, Hyunseung Choo, Khisu Beom, Smartphones as smart pedagogical tools: Implications for smartphones as u-learning devices, Computers in Human Behavior, Volume 27, Issue 6,2011, Pages 2207-2214, ISSN 0747-5632, https://doi.org/10.1016/j.chb.2011.06.017

[37] S. Saraswathi. (2017). Smartphone usage among students. International Education and Research Journal, Vol 3, No 6.

[38] Lopez, Franci Suni. "Actionable emotion detection in context-aware systems." PhD diss., Universidad Católica San Pablo, 2018.

[39] Singh, Anjali and Dinesh Kumar Srivastava. "Understanding the Effect of Internet Addiction on Student Academic Engagement." IJICTE vol.17, no.4 2021: pp.1-12. http://doi.org/10.4018/IJICTE.20211001.oa11

[40] Abayomi Moradeyo Otebolaku, Maria Teresa Andrade, User context recognition using smartphone sensors and classification models,Journal of Network and Computer Applications,Volume 66, 2016, Pages 33-51, ISSN 1084-8045,https://doi.org/10.1016/j.jnca.2016.03.013

[41] Xie H, Peng J, Qin M, Huang X, Tian F, Zhou Z. Can Touchscreen Devices be Used to Facilitate Young Children's Learning? A Meta-Analysis of Touchscreen Learning Effect. Front Psychol. 2018 Dec 18; 9:2580. http://doi.org/10.3389/fpsyg.2018.02580

[42] Christoph Kofler, Martha Larson, and Alan Hanjalic. 2016. User Intent in Multimedia Search: A Survey of the State of the Art and Future Challenges. ACM Comput. Surv. 49, 2, Article 36 (June 2017), 37 pages. https://doi.org/10.1145/2954930

[43] Hon Chu, V. Raman, J. Shen, A. Kansal, V. Bahl and R. R. Choudhury, "I am a smartphone and I know my user is driving," 2014 Sixth International Conference on Communication Systems and Networks (COMSNETS), 2014, pp. 1-8, doi: 10.1109/COMSNETS.2014.6734870.

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:54 UTC from IEEE Xplore. Restrictions apply.